

Chapter-1

Data Abstraction:-

Since many database system users are not computer trained, developers hide the complexity from users through several levels of abstraction to simplify users' interactions with the system.

* Physical Level:- The lowest level of abstraction. Describes how data are actually stored. The physical level describes complex low level data structures in detail.

* Logical Level:-

- Describes what data are stored in database and what relationship exists among those data. Describes database by relatively simple structures.

- Although implementation of simple structures may involve complex physical level structures the user of the logical level does not need to be aware of this complexity.

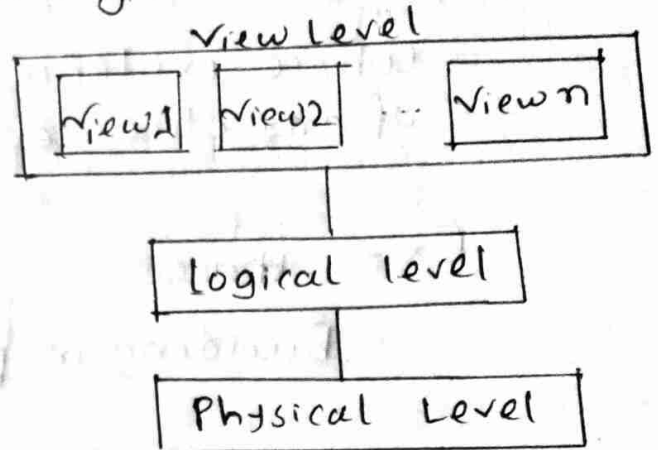
- Data administrator use logical level abstraction.

* View Level:-

- Many users of database system need to access only a part of the database. The view level abstraction exists to simplify their interaction

with the system.

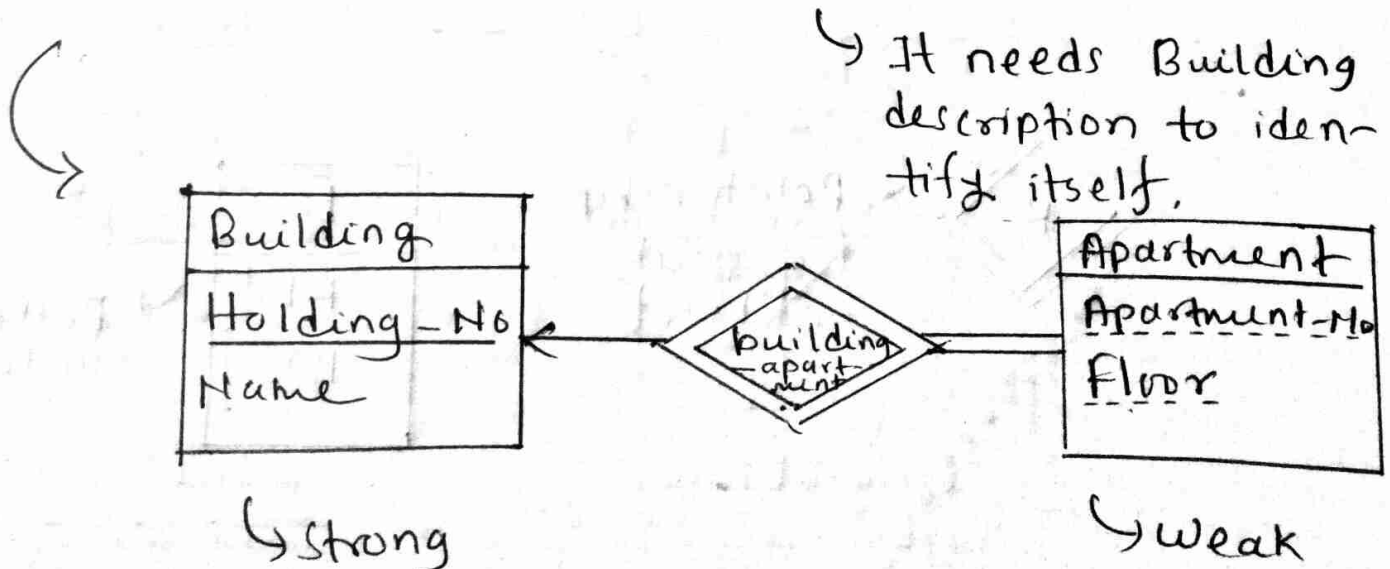
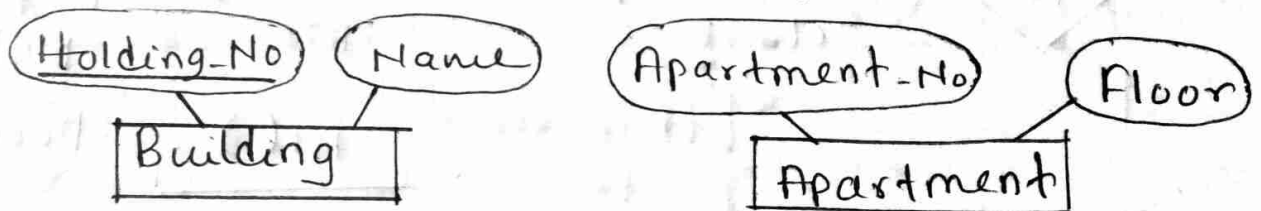
- The system may provide many views for the same database.



Chapter-7:-

Weak Entity Set:-

- Is an entity set that cannot be uniquely identified by its attributes alone. Therefore, it must use a foreign key in conjunction with its attributes to create a primary key.
- The foreign key is typically a primary key of an entity it is related to.



□ Strong Entity set: -

- whose existence does not depend on the existence of any other entity in a schema.

Ex - ~~House~~

Building in previous example

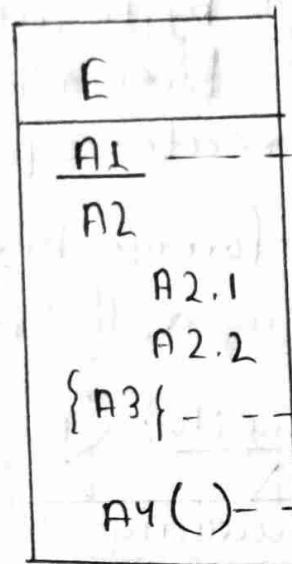
□ ER Notations: -



→ Entity set



→ Relationship set



Primary key

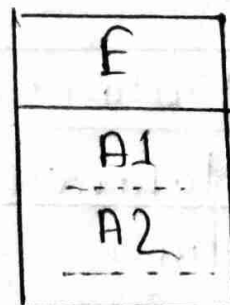
} Composite Attrib

- - - Multivalued

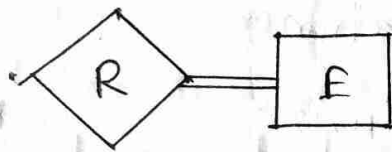
- - - Derived



→ Relationship for weak entity set



→ Dotted underline



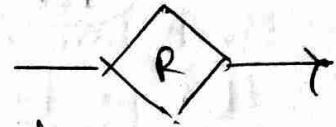
→ Total participation of an entity set in a relation



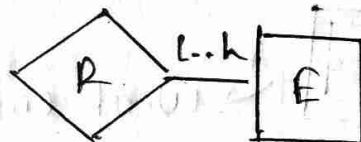
Many to many



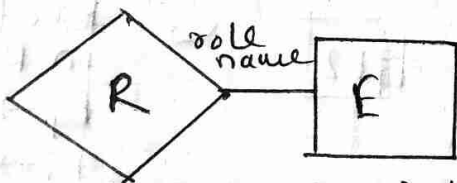
One to one



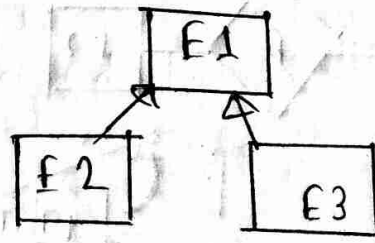
Many to one



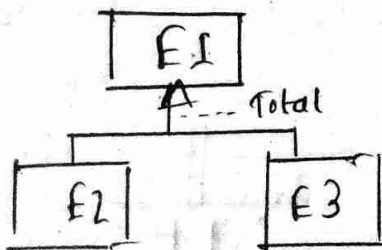
→ Cardinality limits



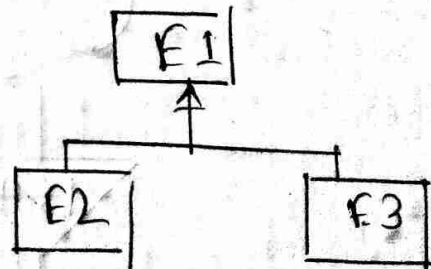
Role indicator



→ ISA: generalization or specialization

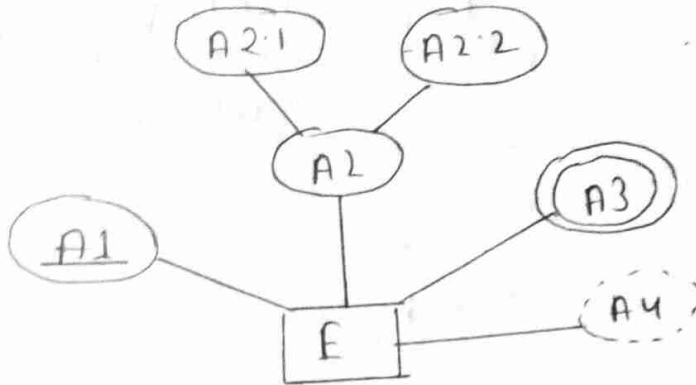


→ Total (disjoint) generalization



→ Disjoint generalization

□ Alternative ER-Notations



A1 → Simple and Primary key

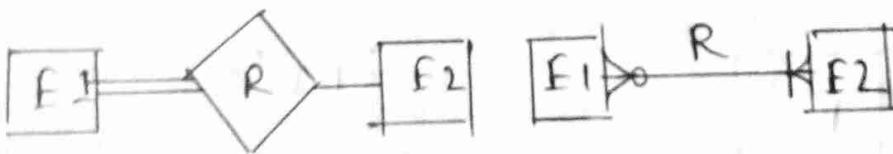
A2 → Composite

A3 → Multivalued

A4 → Derived



→ weak entity set



Participation in R

E₁ → Total

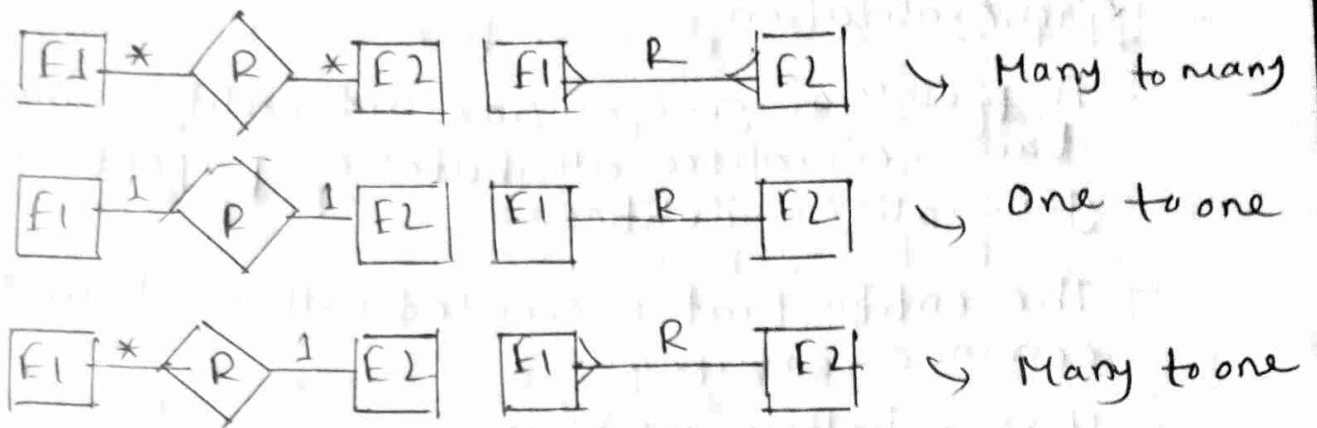
E₂ → Partial



Generalization



Total Generalization

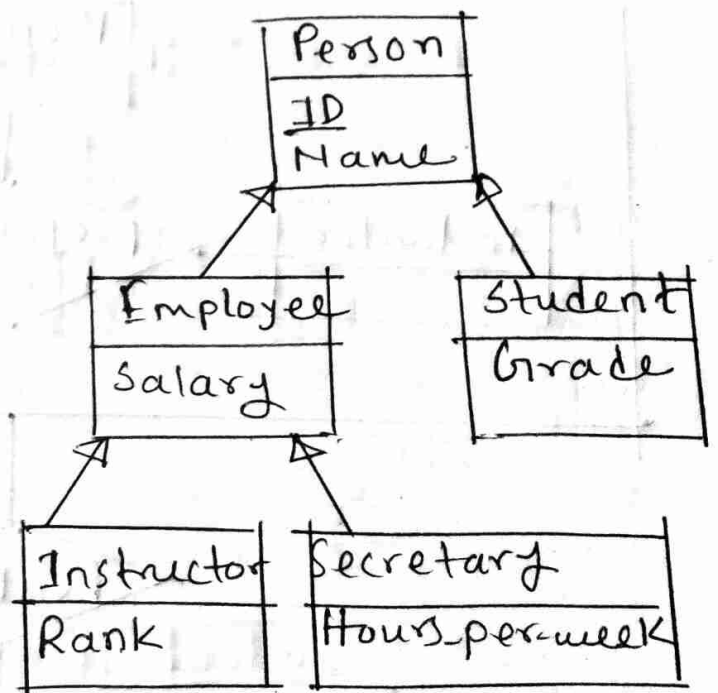


□ Extended ER Features:

* Specialisation

— It is the process of identifying subsets of an entity that shares different characteristics.

— It breaks an entity into multiple entities from higher level (super class) to lower level (sub class).



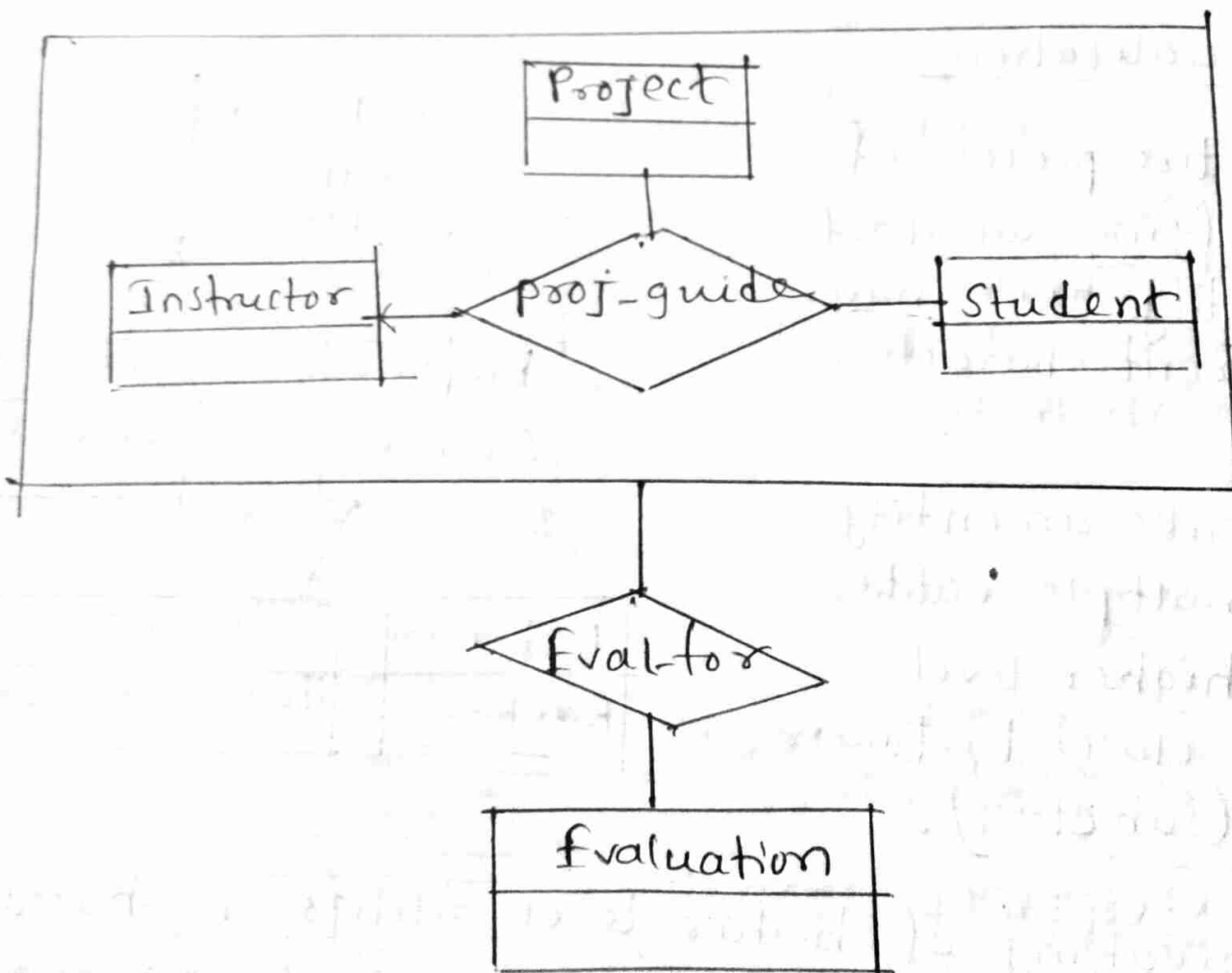
— The breaking of higher level entities is based on some distinguishing characteristics of entities in the super class. It is a top-down approach.

* Generalization:-

- A process of generalizing an entity which contains generalized attributes or properties of generalized entities.
- The entity that is created will contain the common features.
- It is a bottom up process.

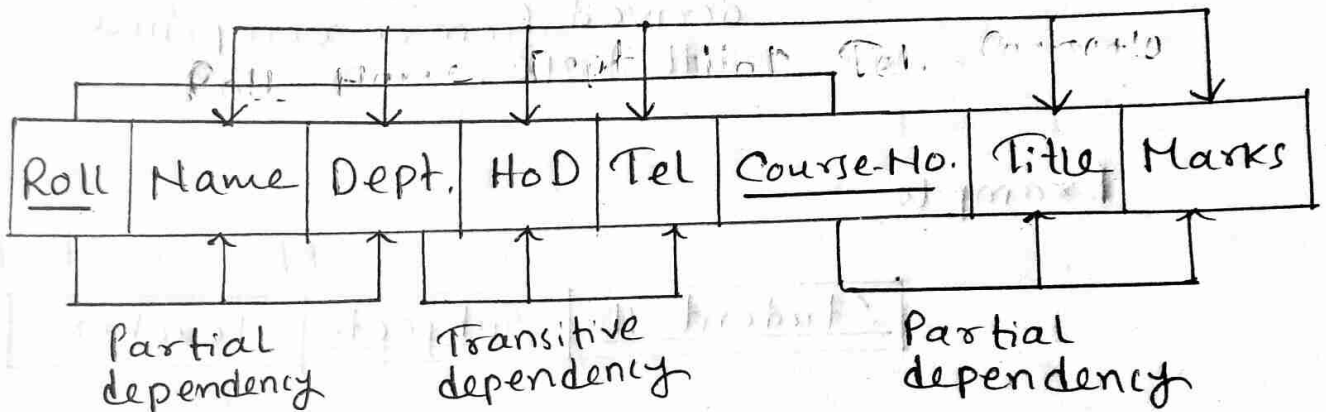
* Aggregation:-

- It is the relationship among relationships.



Chapter-8

□ Dependency Diagram:-



□ Third Normal form (3NF):-

- Satisfies 1NF, 2NF.
- No transitive dependency.

→ Dependency on non-key attribute

Solution ~

Student

<u>Roll</u>	Name	Dept
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Course

<u>Course-No.</u>	Title
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Dept.

<u>Dept</u>	HoD	Tel.
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Stu-Marks

<u>Roll</u>	<u>Course-No.</u>	Marks
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□ Boyce-Codd Normal Form (BCNF 3.5NF) :-

- It should be in 3NF.
- For any dependency $A \rightarrow B$ (B depends on A), A should be a super key.

→ A prime attribute cannot be derived from a non-prime attribute.

Example -

<u>Student-ID</u>	<u>Subject</u>	Teacher
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$(\text{Student-ID}, \text{Subject}) \rightarrow \text{Teacher}$

$\text{Teacher} \rightarrow \text{Subject}$

→ Prime attribute depends on non-prime attribute

Solution



<u>Student-ID</u>	<u>Teacher-ID</u>
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<u>Teacher-ID</u>	Teacher	<u>Subject</u>
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4NF, 5NF → Not for exam, but for real life

Chapter-3

□ Varchar vs. Varchar2 :-

Varchar2 → Does not distinguish between NULL and empty string.

Varchar → Distinguishes between ~~an~~ NULL and empty string.

PL SQL